Francesco Lorenzi

Curriculum vitæ

Personal information

Date of birth	15 September 1998
Nationality	Italian
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Professional experience

October 2025 – September 2026

Postdoctoral researcher, *Photonics and Electromagnetics group, Department of Information Engineering, University of Padova, Italy*
I studied how electromagnetic waves propagate in nonlinear and dispersive media, in particular slow light and spatiotemporal solitons. In parallel, I worked on a problem from fibre telecommunications: the Kerr nonlinearity of the fibre mixes the signals and creates an interference noise that limits the capacity of the link. With analytical methods inspired by wave turbulence theory, I derived how this noise scales and what its statistics are, and developed fast algorithms to estimate it.

October 2022 – September 2025

Doctoral researcher, *Quantum Information and Matter group, Department of Physics and Astronomy, University of Padova, Italy*
My doctorate dealt with nonlinear waves in ultracold quantum gases and optical media. I studied what matter-wave and optical solitons have in common and when they are stable, how low-energy atomic collisions determine the effective interactions in a many-body theory, and the corrections due to fluctuations beyond the mean-field approximation in bosonic Josephson junctions and in Bose gases in general, combining analytical techniques, such as variational approaches and two-body scattering theory, with the numerical integration of the Gross–Pitaevskii and related equations. Some of the results were directly compared with a cold-atom experiment. Part of the research was devoted to the coupled dynamics of the electromagnetic field in an optical cavity and a moving mirror, investigated by means of a variational method.

January 2022 – October 2022

Research collaborator, *Photonics and Electromagnetics group, Department of Information Engineering, University of Padova, Italy*
Before the doctorate I worked on the theory and modelling of nonlinear light propagation in optical fibre. I used a perturbative approach for modelling the impact of fibre nonlinearity on realistic telecommunications systems, and explored the same approach for quantum key distribution channels carried alongside classical traffic.

Education

March 2026

Ph.D. in Physics, *University of Padova, Italy*
Supervisor: Prof. Luca Salasnich.
Thesis title: *Nonlinear physics of ultracold quantum gases and optical fields in media*, defended *cum laude*.

September 2022

M.Sc. in ICT (photonic engineering track), *University of Padova, Italy*
Supervisor: Prof. Marco Santagiustina.
Thesis title: *Nonlinear noise in WDM systems: study of classical and quantum channel interaction and capacity*, final grade 110/110 *cum laude*. Weighted exam average: 29.6/30.

July 2020

B.Sc. in Information Engineering, *University of Padova, Italy*
Thesis title: *Dispersione cromatica e non linearità in fibra ottica, un'analisi simulativa*, final grade 110/110 *cum laude*.

Research visits

October 2023 – February 2024

Barcelona Quantum Monte Carlo group, Universitat Politècnica de Catalunya, *Barcelona, Spain*
Five months as a visiting researcher with Prof. Pietro Massignan, working on scattering theory and effective interactions for ultracold atoms, and on quantum Monte Carlo methods in collaboration with Prof. Grigori Astrakharchik.

Awards and funding

2022–2025 Full three-year Ph.D. scholarship, University of Padova, ≈€52,000 gross in total.

2021–now Teaching assistantship and tutoring contracts for undergraduate physics and mathematics courses, University of Padova, ≈€3,600 gross in total.

2021 "Premio di studio del Rotary Club di Padova", named after Prof. Carlo Giacomo Someda, awarded for academic excellence and service to the community, ≈€1,200.

Publications

Preprints:

A. Bardin, **F. Lorenzi** and L. Salasnich, *Bosonic Josephson junction dynamics: interplay between quantum and thermal fluctuations*, arXiv:2604.27809 (2026).

Journal articles:

12. **F. Lorenzi** and L. Salasnich, *Dimensional reduction for optical beams with thermal nonlocal nonlinearity*, Front. Phys. **21**, 102203 (2026), in the special issue *Topological Dynamics in Ultracold Atoms and Optics*.
11. **F. Lorenzi**, G. Marcon, L. Schenato, A. Mecozzi, C. Antonelli and M. Santagiustina, *Fast estimation of nonlinear interference noise in distributed Raman-amplified few-mode fiber links*, J. Lightwave Technol. **44**, 4476–4490 (2026).
10. **F. Lorenzi**, M.G. Pelizzo and L. Salasnich, *Variational treatment of the electromagnetic field in an optical cavity with a moving mirror*, J. Phys. Photonics **8**, 015023 (2026).
9. R. Cruickshank, **F. Lorenzi**, A. La Rooij, E.F. Kerr, T. Hilker, S. Kuhr, L. Salasnich and E. Haller, *Experimental observation of single- and multisite matter-wave solitons in an optical accordion lattice*, Phys. Rev. Lett. **135**, 263404 (2025). Selected as Editors' Suggestion, and featured in APS *Physics*: P.G. Kevrekidis, *An accordion lattice playing a soliton tune*, Physics **19**, 3 (2026).
8. **F. Lorenzi**, L. Salasnich and M.G. Pelizzo, *Optical cavity in the relativistic regime for laser propulsion*, Phys. Rev. Applied **24**, 034033 (2025).
7. **F. Lorenzi** and L. Salasnich, *Low-energy atomic scattering: s-wave relation between the interaction potential and the phase shift*, Ann. Phys. **537**, e00227 (2025).
6. **F. Lorenzi** and L. Salasnich, *Variational approach to multimode nonlinear optical fibers*, Nanophotonics **14**, 805–813 (2025).
5. L. Salasnich, M.G. Pelizzo and **F. Lorenzi**, *Only-phase Popov action: thermodynamic derivation and superconducting electrodynamics*, J. Phys. A: Math. Theor. **57**, 355302 (2024).
4. **F. Lorenzi** and L. Salasnich, *Atomic soliton transmission and induced collapse in scattering from a narrow barrier*, Sci. Rep. **14**, 4665 (2024).
3. A. Bardin, **F. Lorenzi** and L. Salasnich, *Quantum fluctuations in atomic Josephson junctions: the role of dimensionality*, New J. Phys. **26**, 013021 (2024).
2. **F. Lorenzi**, G. Marcon, A. Galtarossa, L. Palmieri, A. Mecozzi, C. Antonelli and M. Santagiustina, *Nonlinear interference noise in Raman-amplified WDM systems*, J. Lightwave Technol. **41**, 6465–6473 (2023).

1. **F. Lorenzi**, A. Bardin and L. Salasnich, *On-shell approximation for the s-wave scattering theory*, Phys. Rev. A **107**, 033325 (2023).

Proceedings:

F. Lorenzi, L. Schenato, A. Mecozzi, C. Antonelli and M. Santagiustina, *Time-domain two-pulse collisions nonlinear interference noise model for ultra-wideband links*, ECOC 2026, Málaga, Spain, 20–24 September 2026.

F. Lorenzi and L. Salasnich, *Dimensional reduction for paraxial beams with thermally mediated nonlocal nonlinearity*, in the proceedings of RiNEM 2026, L'Aquila, Italy, 6–9 September 2026.

F. Lorenzi and M. Santagiustina, *Beer–Lambert absorption enhancement in structural and material slow-light*, in the proceedings of RiNEM 2026, L'Aquila, Italy, 6–9 September 2026.

F. Lorenzi, G. Marcon, A. Galtarossa, L. Palmieri, A. Mecozzi, C. Antonelli and M. Santagiustina, *Model for nonlinear interference noise in Raman-amplified WDM systems*, in the proceedings of the ECOC 2022, paper Tu1D.4.

Book:

L. Salasnich and **F. Lorenzi**, *Modern Physics: An Invitation to Statistical and Quantum Mechanics*, UNITEXT for Physics, Springer Nature Switzerland, 2nd edition (2025). Italian edition: *Introduzione alla fisica moderna*, Springer (2026).

Talks and posters

September 2026 — "*Dimensional reduction for paraxial beams with thermally mediated nonlocal nonlinearity*" and "*Beer–Lambert absorption enhancement in structural and material slow-light*". Talks at RiNEM 2026, L'Aquila, Italy.

June 2026 — "*Fast estimation of nonlinear interference noise in distributed Raman-amplified few-mode fiber links*". Talk at ICOP 2026, L'Aquila, Italy.

January 2026 — "*Experimental observation of single- and multisite matter-wave solitons in an optical accordion lattice*". Poster at the QSnow–QTech workshop, Asiago, Italy.

August 2025 — "*Variational approach to multimode nonlinear optical fibers*". Poster and flash presentation at the Bad Honnef Physics School on Ultracold Quantum Matter, Bad Honnef, Germany.

May 2024 — Poster on quantum fluctuations in atomic Josephson junctions, co-presented with A. Bardin, at the Atomtronics conference, Benasque, Spain.

January 2024 — "*Scattering of matter-wave soliton from a narrow barrier: transmission coefficient and induced collapse*". Poster at the CAPS Cold Atom Workshop, Barcelona, Spain.

September 2023 — "*Universality and effective range corrections in quantum gases*". Talk at the CMD30/FisMat 2023 conference, Milan, Italy.

September 2022 — "*Model for nonlinear interference noise in Raman-amplified WDM systems*". Talk at ECOC 2022, Basel, Switzerland.

A collection of presentations, seminars and posters is available at lorenzifrancesco.github.io.

Peer review activity

I have reviewed manuscripts for *Nature Communications*, *Physics Letters A*, *Journal of Low Temperature Physics*, *Journal of Lightwave Technology* and *Scientific Reports*.

Teaching experience

2021–now — Teaching assistant for the courses "Fisica 1", "Fisica 2" and "Fondamenti di algebra lineare e geometria", University of Padova.
Supplementary teaching in the physics laboratories of the engineering degrees.

2021–2024 Tutor for the university inclusion programme, for the courses "Analisi 1" and "Consapevolezza digitale", University of Padova.

February 2021 Information technology teacher, Istituto Tecnico Tecnologico Economico (ITTE) G. Galilei, Arzignano (VI), Italy.

Other activities

2025–now Co-organiser of the FNIP conference (*Focus on Optical NeuroImaging and Photonics*), University of Padova: responsible for theme T3, *simulations and numerical methods*, and member of the organising committee for FNIP day 2026.
Main organiser of the 9th FNIP webinar, T3 webinar "Open-source tools for classical and quantum optics" (October 2025).

2025 Co-organiser of PRAPI (*Ph.D. Research Activity Presentation Initiative*), a seminar series in which doctoral students present their research in accessible terms to an audience from other fields of physics, Department of Physics and Astronomy, University of Padova, Italy.

2024–2025 Representative of the Ph.D. students in the Consiglio direttivo of the doctoral school, Department of Physics and Astronomy, University of Padova, Italy.

July 2022 Designer and teacher of the digital literacy course "Strumenti digitali per la cittadinanza" for the general public, Biblioteca comunale G. Bedeschi, Arzignano (VI), Italy.
The course was accompanied by a study on the digital divide based on data from the participants.

Languages

Italian Native.

English Fluent, working language for research and teaching.
Certified Cambridge FCE B2 in February 2016.

German, Spanish and French Basic reading comprehension.

Computer and technical skills

Advanced `Python`, `Julia`, `MATLAB`, `Git/GitHub`, LaTeX, and running simulations on `SLURM`-managed clusters.
Linux is my operating system of choice, and I am confident with computer maintenance, both software and hardware.

Intermediate `C`, `C++`, `Rust`, `CST-FEM`.

Basic `Fortran`, `x86` assembly, embedded programming on `STM32` (ARM) microcontrollers, `SQL`.

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updated November 2026.